ICTCE04

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Weekly Progress Report Form

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Company Name /Employer Name - **HTZKKL Technologies Com.,Ltd.**

Reporting Period: *Week No. 1, Month 2 - 9/9/2026, Total 40 Hours*

1. Name of project/task/issue/topic of student's responsibility.

(List each of the performed tasks.)

Project: Multi-Tenant School Intranet & Academic Management Platform

Task 1: Git repository cloning, local development environment setup (FastAPI + Next.js + PostgreSQL), and review of developer handover documentation (HANDOVER.md).

Task 2: Architecture design and implementation of the Multi-Tenant Engine (v2.0) with school-level data isolation and dynamic feature toggles.

Task 3: Security audit and hardening: removing hardcoded bootstrap credentials, deleting debug scaffolding endpoints, and refactoring JWT authentication decoding helpers.

Task 4: Development of the Parent Portal module (parent-to-student relationship mapping and /dashboard/my-children portal).

Task 5: Implementation of the Bulk CSV onboarding engine for students/teachers and automated verification script development.

2. Summary of the performed tasks.

(Describe the type of work that the student has accomplished.)

During the first week of starting the project:

1.Repository Cloning & Initial Setup: Cloned the project repository, configured

PostgreSQL database connections, configured Python virtual environment with FastAPI backend dependencies, and compiled the Next.js frontend. Studied existing Phase 2 codebase architecture and database schema.

2.Multi-Tenancy Transformation (v2.0 Architecture): Migrated the legacy single-school architecture to a true multi-tenant system. Created the School entity, executed Alembic database migrations, and implemented strict school_id data scoping across all entities (students, teachers, classes, and users). Added per-school feature flags to allow schools to toggle modules on demand.

3.Security Audit & Codebase Hardening: Conducted a comprehensive code audit across backend and frontend services:

Eliminated hardcoded fallback passwords from auth/service.py and secured credentials via .env environment variables (BOOTSTRAP_ADMIN_PASSWORD).

Removed exposed debug routes (GET /auth/admin-only).

Refactored redundant base64 JWT decoding blocks in frontend/components/auth.ts into a unified, secure decodeTokenPayload() helper.

4.Parent Portal & Bulk Data Ingestion:

Implemented the Parent entity and student-parent relational mapping, enabling parents to log in and securely view attendance, timetables, and academic reports for their children.

Built a bulk CSV import engine with database transaction savepoints to onboard student and teacher rosters in bulk.

5.Verification & Testing Suite: Authored standalone end-to-end verification scripts (verify_school_feature_toggle.py,verify_parent_module.py,verify_bulk_csv_module.py, verify_analytics_module.py) and updated project architectural documentation in the system vault.

3. Application of concepts, theories, knowledge, or experiences that applied to cooperative education.

Software Architecture & Multi-Tenancy: Applied database multi-tenancy principles (shared database, tenant discriminator column school_id) to ensure secure data isolation across separate school organizations.

Database Design & Relational Modeling: Utilized SQLAlchemy ORM and Alembic migration tools to design relational foreign keys, cascade rules, and indexes for parent-student linkages and school settings.

Application Security & Authentication: Applied principle of least privilege, secure

credential management (environment variables vs. hardcoding), and stateless authentication using JSON Web Tokens (JWT).

Defensive Programming & Transaction Safety: Implemented database transactions with rollback/savepoint handling (db.begin_nested()) during CSV parsing to prevent partial corrupted data ingestion.

DevOps & Documentation: Maintained structured project documentation, architecture decision records (ADRs), and Git feature branching workflows (feature/v2.0-multi-tenant-system).

4. Problems/obstacles encountered during the work.

(Provide details on how to solve the problems/obstacles.)

Problem 1: Cross-Tenant Data Leakage Risk in Existing Single-Tenant Queries:

Detail: The initial codebase assumed a single school, meaning existing queries did not filter by school_id. There was a critical risk that one school's administrators could query or modify records belonging to another school.

Solution: Inject mandatory school_id scoping in backend route dependencies and service layers (verify_class_access, verify_student_access), ensuring any resource requested outside the user's assigned school returns a 404 Not Found or 403 Forbidden.

Problem 2: Hardcoded Secrets and Inefficient Client-Side JWT Decoding:

Detail: Found hardcoded fallback administrative passwords in source code and 4 duplicate implementations of base64 token parsing on the frontend, which caused maintenance and security concerns.

Solution: Refactored configuration loading to strictly require BOOTSTRAP_ADMIN_PASSWORD from .env in production, and extracted a single DRY helper decodeTokenPayload() in components/auth.ts with error handling for malformed tokens.



(……Nay Min.....)

Job Supervisor

Date ……9/9/2026.....



(….Sai Swam Wan Hline.....)

Student

Date ……9/9/2026.....